

Amazon Delivery Time Prediction Using Machine Learning

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Project Type: Python + Machine Learning Internship Project

Introduction

With the rapid growth of online shopping, delivery time prediction has become crucial for customer satisfaction and operational efficiency. This project focuses on predicting Amazon parcel delivery time (in hours) based on features such as agent age, rating, distance, weather, traffic, and delivery category.

Objective

To build a machine learning model that accurately predicts the estimated delivery time for Amazon orders based on historical delivery data.

Dataset Description

Dataset: amazon_delivery.csv (~10,000 records). Important columns include Agent_Age, Agent_Rating, Store_Latitude, Store_Longitude, Drop_Latitude, Drop_Longitude, Order_Date, Order_Time, Pickup_Time, Weather, Traffic, Vehicle, Area, Category, and Delivery_Time (target variable).

Data Preprocessing

Performed merging of date and time columns, calculation of pickup delay (Pickup_Lag_min), computation of distance using Haversine formula, extraction of Hour and DayOfWeek, encoding categorical columns numerically, and filling missing values with median.

Exploratory Data Analysis

Visualized delivery time distribution, correlations, and patterns. Observed that longer distances and high traffic increase delivery time, while experienced agents deliver faster.

Model Training

Used Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost. Evaluated with MAE, RMSE, and R^2 metrics. Random Forest performed best with MAE $\approx$ 2.1 hours, RMSE $\approx$ 3.0 hours, $R^2 \approx$ 0.94. Saved as best_model.pkl and metrics.json.

Streamlit Application

Built a user-friendly app for interactive prediction of delivery time based on order details. Users can input parameters like weather, traffic, and distance to get estimated delivery time instantly.

Results

Model comparison table: • Linear Regression — MAE: 3.45, RMSE: 4.21, R^2 : 0.89 • Gradient Boosting — MAE: 2.35, RMSE: 3.14, R^2 : 0.93 • Random Forest — MAE: 2.13, RMSE: 3.01, R^2 : 0.94

Conclusion

Random Forest proved to be the most accurate and stable model. This project demonstrates complete ML workflow including data preprocessing, model training, and deployment using Streamlit.

Future Scope

- Integrate live traffic/weather APIs
- Add real-time route optimization
- Build mobile version of Streamlit app
- Explore deep learning for better accuracy